Aashutosh A V

Atlanta, Georgia
+1(972)-918-1130 avaashutosh04@gmail.com linkedin.com/in/aashutosh-av Portfolio

Education

Georgia Institute of Technology **Aug. 2025 – May 2027**
Master of Science in Computer Science (Specialization: Machine Learning) *GPA: 4.0/4.0*

Birla Institute of Technology & Science, Pilani **Aug. 2021 – Jun. 2025**
Bachelor of Engineering in Computer Science (Minor: Data Science) *GPA: 9.1/10*

Experience

C21U, Georgia Tech **Dec. 2025 – Present**
Graduate Research Assistant (Advisors: Dr. Jeonghyun Lee, Dr. Rod Valente) *Atlanta, GA*

- Developing **autonomous LLM-powered teaching agents** for Socratic Minds, GT's production AI tutoring platform, focusing on human-agent trust calibration and adaptive pedagogical reasoning.
- Designed a **few-shot prompting pipeline** using **GPT-4o** for RAG-assisted ICAP cognitive engagement classification, enabling scalable automated dialog coding across human-AI interaction traces.
- Built a multi-user evaluation framework on **AWS EC2** with **DynamoDB** to log real-time interaction traces; quantified Inter-Rater Reliability (IRR) between human annotators and the LLM agent for **human-in-the-loop** validation.

Hays Lab, Georgia Tech **Dec. 2025 – Present**
Graduate Researcher (Advisor: Dr. James Hays) *Atlanta, GA*

- Building **NVS-Bench**, a unified evaluation framework for sparse-view **Novel View Synthesis** benchmarking diffusion and feedforward models (SEVA, LagerNVS) across context selection strategies on **ScanNet++**.
- Conducting systematic ablations over context-view selection heuristics and intermediate view generation strategies, with interpretability analysis of **temporal attention mechanisms** to diagnose geometric failure modes in state-of-the-art NVS models.

Microsoft Research **Jan. 2025 – July 2025**
Research Intern (Advisor: Dr. Ayush Choure) *Bangalore, India*

- Demonstrated a **35% reduction in cluster resource footprints** while sustaining **99.999% availability** across **100,000+ Azure VM nodes**, projecting **$100M+ annual cost savings** for Azure Cosmos DB.
- Co-invented and filed a **U.S. patent** for ORBIT, a vector-based multi-dimensional online bin-packing algorithm for NoSQL shard replica placement across large-scale distributed infrastructure.
- Engineered **LUNA**, a production time-series forecasting pipeline using **LightGBM quantile regression** and Whittaker-Eilers smoothing to predict peak resource loads under asymmetric loss constraints.
- Formulated **DRV (Distressed Resource Volume)**, a stochastic QoS metric leveraging **Kernel Density Estimation** and probabilistic bootstrapping to model tail-error rates against real-time workload intensities.
- Built and open-sourced **LOADSTAR**, a high-fidelity policy simulation framework using non-parametric statistical models to benchmark placement algorithms on production-scale traces; enabled rapid offline ML policy evaluation before live deployment, reducing multi-tenant compute variance ($\ln(P80/P20) \to 1.34$).

Publications & Patents

ScenarioCLIP: Visual Language Models for Natural Scene Analysis | *WACV (Core A), Under Review* **2027**
- Proposed a **relation-aware VLM** with disentangled encoders and EMA-based intra-modal knowledge distillation for compositional scene understanding; curated the **Action-Genome Dataset** (1.8M image-text pairs).

Analyzing and Optimizing NoSQL Workloads for Cosmos DB | *PVLDB Vol. 19 (Core A*)* **2026**
- Characterized NoSQL workload dynamics at datacenter scale, deriving analytical models for tail-error prediction and replica placement optimization across **100,000+ node** Azure Cosmos DB clusters.

Narrating For You: Audio-visual Narrating Face Generation | *WACV 2026 (Core A)* **2026**
- Introduced a **multi-entangled latent space** that jointly encodes text prompts, voice profiles, and facial identity to enable synchronized **joint audio-video generation** of talking faces from a single static image.

Workload Distribution Based on Projected Error Counts | *U.S. Patent & Trademark Office (Microsoft)* **2025**
- Novel error-projection scheme that preemptively redistributes NoSQL workloads based on **forecasted error counts**, preventing QoS violations before they occur across distributed database clusters.

Latent Flow Diffusion for Deepfake Video Generation | *CVPR 2024 (Workshop)* **2024**
- Proposed **LFD2G**, a three-stage deepfake generation pipeline combining latent **optical flow prediction**, dual Vision Transformer encoding, and flow-conditioned diffusion to disentangle spatial identity from temporal dynamics.

Technical Skills & Achievements

Languages: Python, C/C++, SQL, Bash.
AI/ML Frameworks: PyTorch, JAX, Hugging Face, OpenCV, LightGBM, Scikit-Learn, MLflow, W&B.
Systems & Tools: Amazon AWS (EC2, DynamoDB), Azure, Distributed Systems, Git, HPC.